

PART 4: SEQUENTIAL LOGIC

DESIGN PROCEDURE OF SEQUENTIAL CIRCUITS

SAAD BAIG – DIGITAL LOGIC DESIGN



The animations and contents in these slides are originally adapted from the DLD course slides designed by Dr. Hasan Baig and Mr. Junaid Ahmed Memon. The original slides can be accessed from <https://www.hasanbaig.com/resources>

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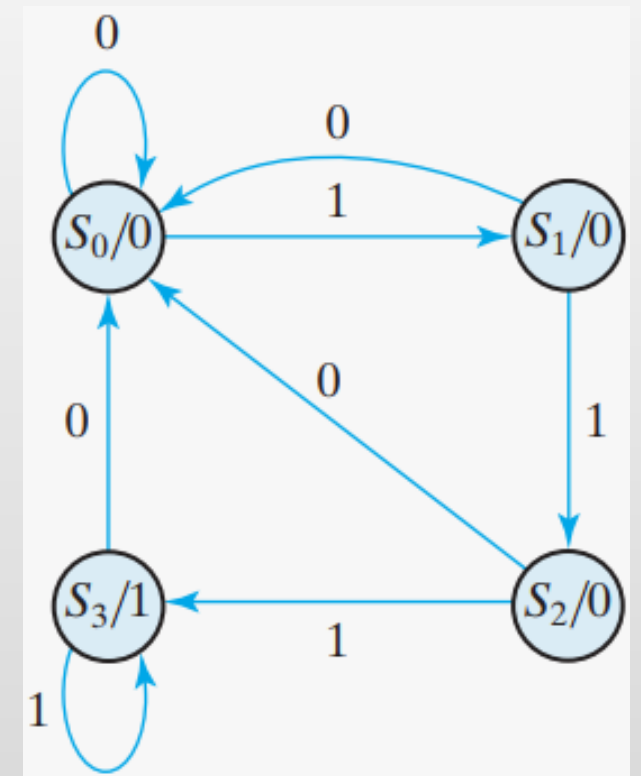


DESIGN PROCEDURE

- The design of a clocked sequential circuit starts from a set of specifications and culminates in a logic diagram or a list of Boolean functions from which the logic diagram can be obtained.
- Sequential circuit design requires a state table or equivalent for its specification.
- The design of the circuit consists of choosing the flip-flops and then finding a combinational gate structure that, together with the flip-flops, produces a circuit which fulfills the stated specifications.
- The number of flip-flops is determined from the number of states needed in the circuit and the choice of state assignment codes.
- Once the type and number of flip-flops are determined, the design process involves a transformation from a sequential circuit problem into a combinational circuit problem. In this way, the techniques of combinational circuit design can be applied.

DESIGN PROCEDURE EXAMPLE

- Problem Statement: Design a logic circuit which detects a sequence of three or more consecutive 1's in a string of bits coming through an input line (i.e. the input is serial bit stream).
- I. From the word description and specifications of the desired operation, derive a state diagram for the circuit.

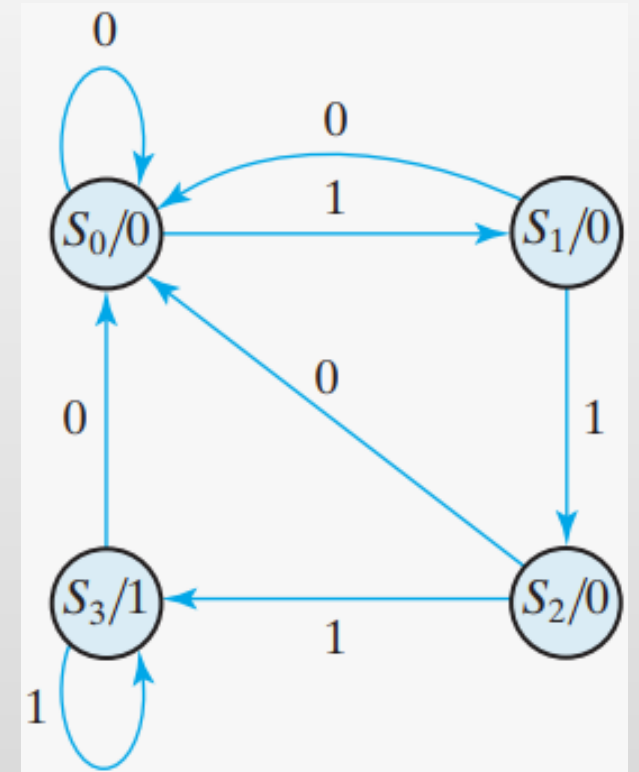


DESIGN PROCEDURE EXAMPLE

2. Reduce the number of states if necessary.
3. Assign binary values to the states
4. Obtain the binary-coded state table

State Table for Sequence Detector

Present State		Input x	Next State		Output y
A	B		A	B	
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	0	0
0	1	1	1	0	0
1	0	0	0	0	0
1	0	1	1	1	0
1	1	0	0	0	1
1	1	1	1	1	1





DESIGN PROCEDURE EXAMPLE

5. Choose the type of flip-flops to be used.
 - Let's assume D-FF.
6. Derive the simplified flip-flop input equations and output equations.

State Table for Sequence Detector

Present State		Input <i>x</i>	Next State		Output <i>y</i>
<i>A</i>	<i>B</i>		<i>A</i>	<i>B</i>	
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	0	0
0	1	1	1	0	0
1	0	0	0	0	0
1	0	1	1	1	0
1	1	0	0	0	1
1	1	1	1	1	1

$$A(t + 1) = D_A(A, B, x) = \Sigma(3, 5, 7)$$

$$B(t + 1) = D_B(A, B, x) = \Sigma(1, 5, 7)$$

$$y(A, B, x) = \Sigma(6, 7)$$

DESIGN PROCEDURE EXAMPLE

5. Choose the type of flip-flops to be used.
 - Let's assume D-FF.
6. Derive the simplified flip-flop input equations and output equations.

		B			
		00	01	11	10
A	0	m_0	m_1	1	m_2
	1	m_4	1	1	m_6

x
 $D_A = Ax + Bx$

		B			
		00	01	11	10
A	0	m_0	1	m_3	m_2
	1	m_4	1	1	m_6

x
 $D_B = Ax + B'x$

		B			
		00	01	11	10
A	0	m_0	m_1	m_3	m_2
	1	m_4	m_5	1	1

x
 $y = AB$

$$A(t + 1) = D_A(A, B, x) = \Sigma(3, 5, 7)$$

$$B(t + 1) = D_B(A, B, x) = \Sigma(1, 5, 7)$$

$$y(A, B, x) = \Sigma(6, 7)$$

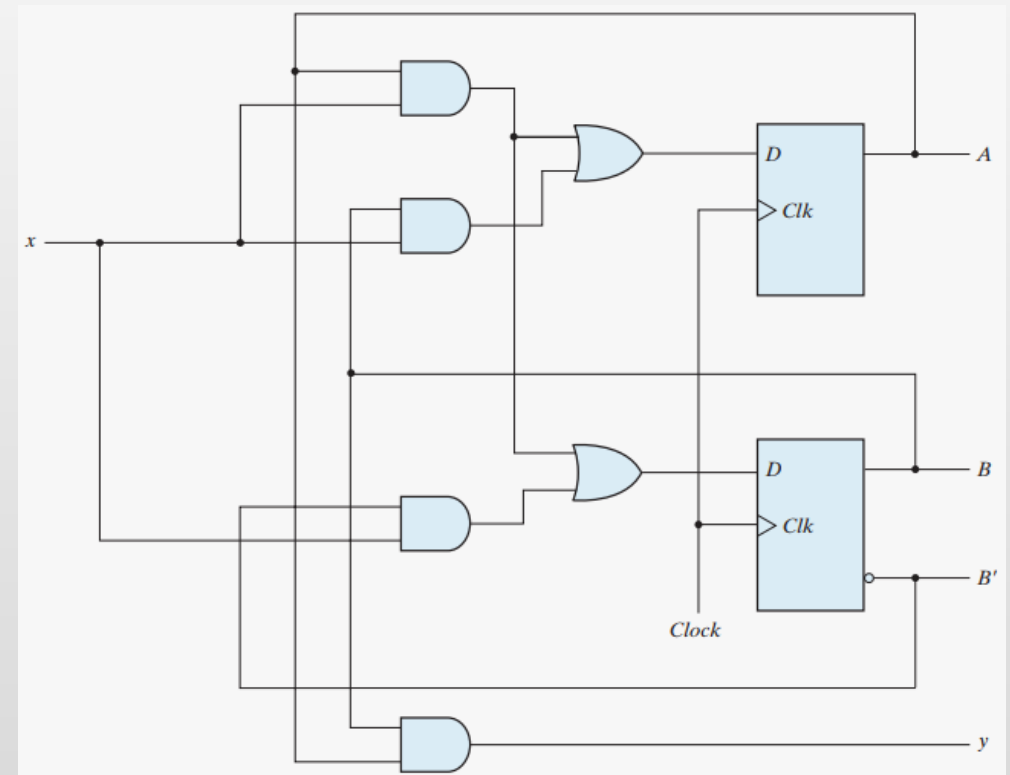
DESIGN PROCEDURE EXAMPLE

7. Draw the logic diagram.

$$D_A = Ax + Bx$$

$$D_B = Ax + B'x$$

$$y = AB$$





DESIGN PROCEDURE SUMMARY

1. From the word description and specifications of the desired operation, derive a state diagram for the circuit.
2. Reduce the number of states if necessary.
3. Assign binary values to the states.
4. Obtain the binary-coded state table.
5. Choose the type of flip-flops to be used.
6. Derive the simplified flip-flop input equations and output equations.
7. Draw the logic diagram.

EXCITATION TABLES

- When D-type flip-flops are employed, the input equations are obtained directly from the next state. This is not the case for the JK and T types of flip-flops.
- During the design process, we usually know the transition from the present state to the next state and wish to find the flip-flop input conditions that will cause the required transition.
- Excitation table is the one that lists the inputs for a given change of state.

J	K	$Q(t + 1)$	
0	0	$Q(t)$	No change
0	1	0	Reset
1	0	1	Set
1	1	$Q'(t)$	Complement

$$Q(t + 1) = JQ' + K'Q$$

$Q(t)$	$Q(t + 1)$	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

JK-FF Excitation Table

T	$Q(t + 1)$	
0	$Q(t)$	No change
1	$Q'(t)$	Complement

$$Q(t + 1) = TQ' + T'Q$$

$Q(t)$	$Q(t + 1)$	T
0	0	0
0	1	1
1	0	1
1	1	0

T-FF Excitation Table

DESIGN PROCEDURE EXAMPLE

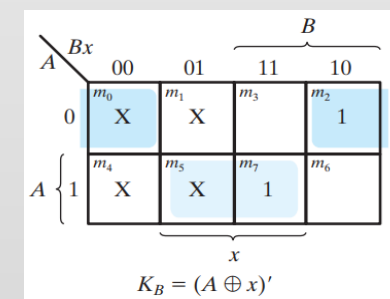
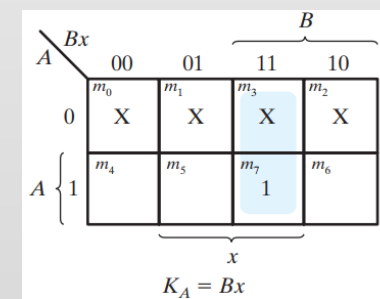
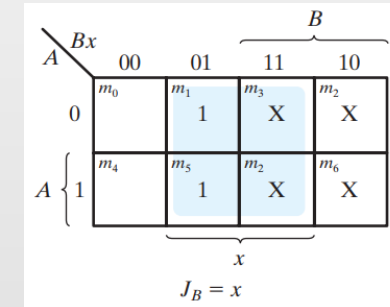
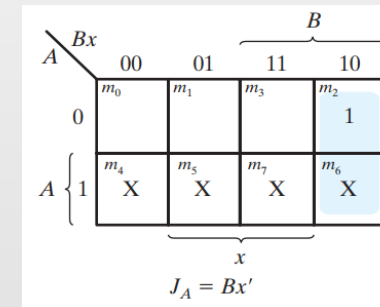
- Design a logic circuit (using J-K flip-flops) which detects a sequence of three or more consecutive 1's in a string of bits coming through an input.

State Table and JK Flip-Flop Inputs

Present State		Input	Next State	
A	B		A	B
0	0	0	0	0
0	0	1	0	1
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	1	1
1	1	0	1	1
1	1	1	0	0

Flip-Flop Inputs

J_A	K_A	J_B	K_B
0	X	0	X
0	X	1	X
1	X	X	1
0	X	X	0
X	0	0	X
X	0	1	X
X	0	X	0
X	1	X	1



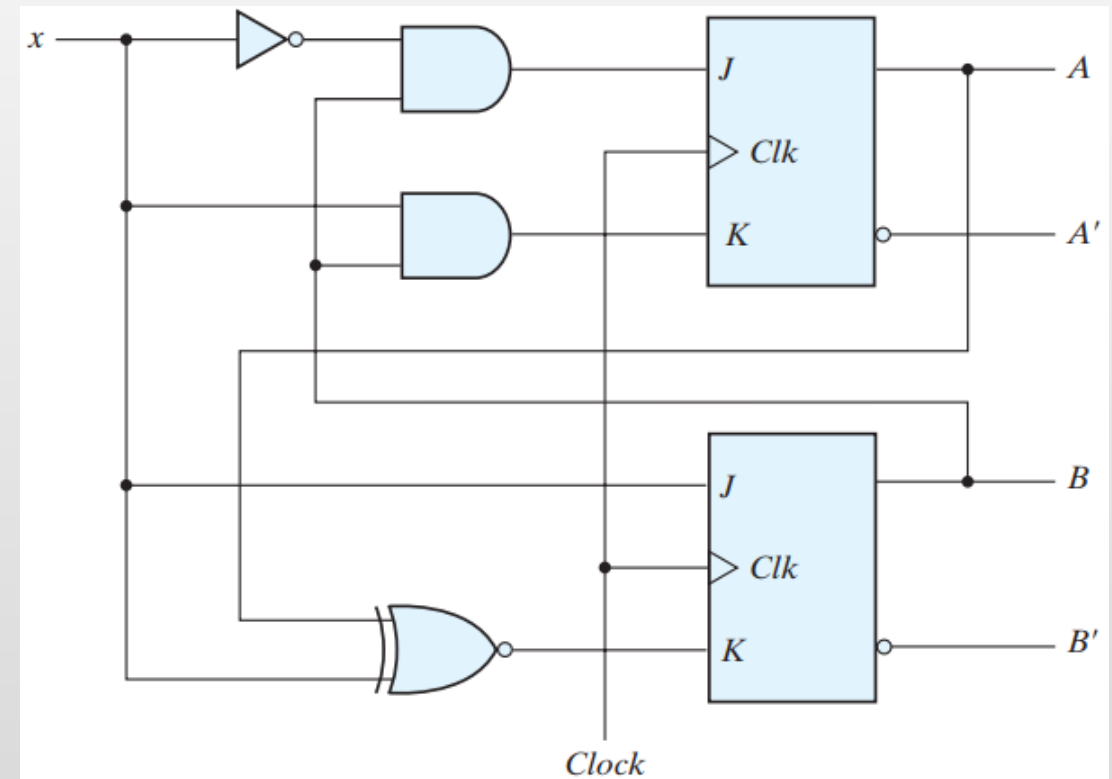
DESIGN PROCEDURE EXAMPLE

$$J_A = Bx'$$

$$K_A = Bx$$

$$J_B = x$$

$$K_B = (A \oplus x)'$$



ACTIVITY

- Design a three bit binary counter using T-FF. The state diagram of the design is show.

